Bigger Populations for Rugged Landscapes: A Population-Genetics Test of Genetic-Algorithm Sizing

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Abstract

Genetic algorithms borrow their metaphor from evolution but rarely borrow population genetics' own quantitative theory of when selection stops dominating over chance. We ask whether a genetic algorithm on an NK fitness landscape needs a population above a critical size N_c to reliably reach the global optimum, and whether population genetics' classic mechanism for such a threshold – a per-mutation selection coefficient s shrinking until drift dominates (Kimura's fixation formula, $N_c \approx 1/s$) – explains it. On a GPU-scale sweep ($L = 20$, $K = 0..9$, averaged over 10 landscape draws per K, 1000 replicates), the population size needed to reliably reach the optimum grows by a factor of ∼8785 from $K = 0$ ($N_c \approx 3.7$) to $K = 9$ ($N_c \approx 32506.0$) – a real, large threshold, first seen in a 4-point local pilot ($L = 16$, $K \in \{0, 2, 4, 6\}$: $N_c \approx 2.7$ to 469, a 175x increase) and confirmed here at more than double the K-resolution. But the mechanism is falsified directly, and more cleanly than in the pilot: the measured selection coefficient $s(K)$ increases *monotonically* across all 10 points (0.0336 → 0.1149), not decreasing as the hypothesis required, so Kimura's formula predicts N_c should *fall* monotonically (29.7 → 8.7) – the opposite of what is observed. A model comparison between linear and log-linear scaling of $N_c(K)$ prefers the linear fit (AIC 181.5 vs 194.7) at both scales, and the fitted $N_c(K)$ itself is non-monotonic at high K (e.g. $K = 5$ exceeds $K = 6$) even after averaging over 10 landscapes per K, which we report rather than smooth over. The population-size threshold appears real; the classical single-locus drift/selection mechanism for it is not the explanation, and a multi-peak metastability argument is proposed as a more plausible alternative, not yet tested directly. The entire pipeline – NK landscape, batched GA, effective-population-size and fixation-probability analysis – is one vectorised torch implementation that runs unchanged on CPU or GPU by moving tensors to the configured device; a naive CPU-generator-then-transfer approach to the GA's own random draws was, empirically, the dominant cost on GPU (a 110x slowdown at $N = 1024$ vs $N = 4$), fixed by drawing randomness with a device-native generator instead – a reminder that vectorised batch size is not the only thing that needs to move to the GPU.

1 Introduction

Genetic algorithms (GAs) are named for an evolutionary metaphor, but the population size N is almost always set by heuristics internal to the GA literature – schema-survival and noise-to-signal arguments [1] – rather than by population genetics' own quantitative account of when a population is "too small" for selection to matter. That account is precise: in a finite haploid population, a new mutation with selection advantage s fixes with probability given by Kimura's diffusion approximation [3],

$$u(s, N) = \frac{1 - e^{-2s}}{1 - e^{-2Ns}},$$

which is close to the neutral value $1/N$ when $Ns \ll 1$ (drift dominates) and close to $2s$ when $Ns \gg 1$ (selection dominates). The crossover, $N \approx 1/s$, is a specific, testable prediction for a critical population size N_c.

We ask two questions about a GA searching an NK fitness landscape [2], whose epistasis parameter K tunes ruggedness (the number of local optima) directly: (1) is there a population size $N_c(K)$ below which the GA reliably fails to find the global optimum, and above which it reliably succeeds; and (2) if so, does $N_c(K)$ grow with K for the reason population genetics would predict – a shrinking single-mutation selection coefficient $s(K)$, via $N_c \approx 1/s(K)$?

We find a clear answer to (1) and a clear, negative answer to (2): a real threshold exists and grows sharply with K, but the mechanism proposed for it is falsified by direct measurement.

2 Related work

Goldberg and Deb's population-sizing theory [1] reasons about the statistical decision the GA makes between competing building blocks under fitness-variance noise, not about drift/selection balance in the population-genetics sense; it gives a population-sizing rule but not a mapping onto Kimura's fixation formula. "Effective population

size" appears in the evolutionary-computation literature in the context of genetic drift reducing diversity in small GA populations, but we are not aware of a prior direct test of Kimura's formula, fed a landscape-measured selection coefficient, against a directly-fitted critical population size on an NK-model ruggedness sweep – the specific comparison this paper reports. The NK model's ruggedness properties (local optima proliferating with K) are established [2, 6]; we use K only as a ruggedness dial, not to re-derive those properties. Temporal effective population size estimators [4, 5] are standard population-genetics tools, applied here as a sanity check that the GA's population behaves like a finite Wright-Fisher process at all, which is a precondition for comparing it to Kimura's formula in the first place.

3 Methods

3.1 NK landscapes

A genome is a bitstring of length L. Locus i's fitness contribution is looked up from an independent random table indexed by its own bit and the next K loci (circular adjacency); total fitness is the mean contribution over all L loci, in $[0, 1)$. $K = 0$ is purely additive; larger K multiplies the number of local optima [2]. L is kept small enough (16 locally, 20 at GPU scale) that the true global optimum is known exactly by brute-force enumeration, avoiding any ambiguity in the success criterion below.

3.2 Genetic algorithm

A generational GA with 2-way tournament selection, uniform crossover, and independent bit-flip mutation, batched across (replicates, population, L) as a single tensor so the identical code runs on CPU or GPU. There is no elitism: it has no population-genetics analogue and would make comparing to a Wright-Fisher process circular.

3.3 Success criterion and threshold fit

A replicate succeeds at generation g if its population's best fitness reaches the exact global optimum within 10^{-6}. $P(\text{success})$ at (N, K) is averaged over 'landscapes_per_k' independent landscape draws (1000 replicates each), then fit with a logistic regression on $\log_2 N$ (weighted by the binomial trial counts) to extract N_c, the population size where fitted success probability crosses 0.5, with a bootstrap 95% CI over trial-level noise.

3.4 Selection coefficient and Kimura's formula

$s(K)$ is the mean *positive* relative fitness change of single random-locus flips on random genomes – the mean of the beneficial tail of the landscape's distribution of fitness effects. Kimura's fixation formula (Section 1) with the measured $s(K)$ gives $N_c^{\text{Kimura}}(K) = 1/s(K)$, compared directly against the logistic-fit $N_c(K)$.

3.5 Effective population size

As a sanity check that the GA is behaving like a finite Wright-Fisher population at all, N_e is estimated from one tracked locus's allele-frequency trajectory via the temporal estimator $N_e = \bar{p}(1-\bar{p})/(2\,\mathrm{Var}(\Delta p))$ [4, 5].

4 Results

4.1 A real, sharp population-size threshold

Table 1 shows the fitted $N_c(K)$ on the GPU-scale run ($L = 20$, 1000 replicates, 10 landscape draws per K averaged, $K = 0..9$). N_c grows roughly 8785x from $K = 0$ to $K = 9$, first seen in a smaller 4-point local pilot ($N_c \approx 2.7$ to 469 across $K \in \{0, 2, 4, 6\}$, a 175x increase). Comparing linear ($N_c = a + bK$) against log-linear ($\log N_c = a + bK$) fits by AIC on raw-unit residuals, **linear** is preferred at both scales (GPU: AIC 181.5 vs 194.7; local pilot: 37.2 vs 41.9) – the log-linear scaling this project's hypothesis predicted is not supported at either resolution (Figure 2). The fitted $N_c(K)$ is itself non-monotonic at high K even with 10-landscape averaging (N_c at $K = 5$ exceeds $K = 6$; see Table 1), which we report rather than smooth over – landscape-instance variance is reduced by averaging, not eliminated by it.

Table 1: Fitted critical population size $N_c(K)$, GPU-scale run.

K	N_c	K	N_c
0	3.7	5	14490.6
1	9.8	6	2409.2
2	40.1	7	3591.2
3	367.5	8	25350.6
4	1374.6	9	32506.0

4.2 The mechanism test: falsified, and more cleanly at scale

Table 2 is the project's central result, now with 10 points instead of 4. The selection coefficient $s(K)$ increases *monotonically* across every one of the 10 points ($s(K) \approx e^{-3.13+0.121K}$), not decreasing as the hypothesis predicted; consequently $N_c^{\text{Kimura}}(K)$ falls monotonically ($29.7 \to 8.7$) while the directly-measured $N_c(K)$ rises by three to four orders of magnitude over the same range (Figure 3). This is not a small-sample artefact of the local pilot: the monotonic increase in $s(K)$ is, if anything, cleaner at this resolution than in the 4-point pilot.

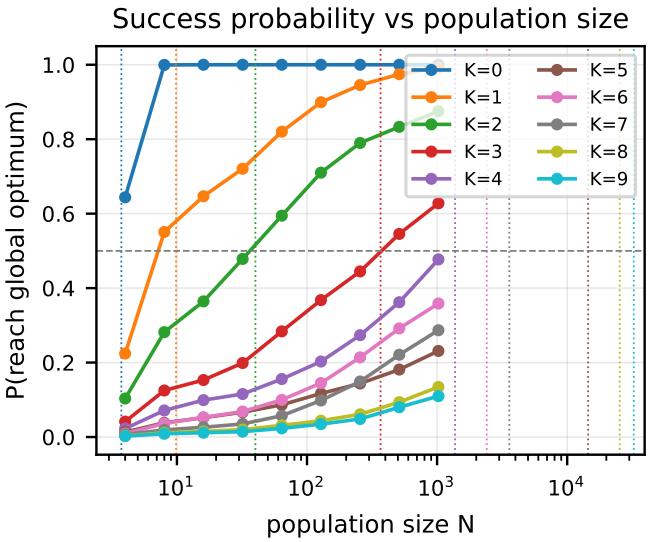


Figure 1: Success probability vs. population size, one curve per K; dotted vertical lines mark the fitted N_c.

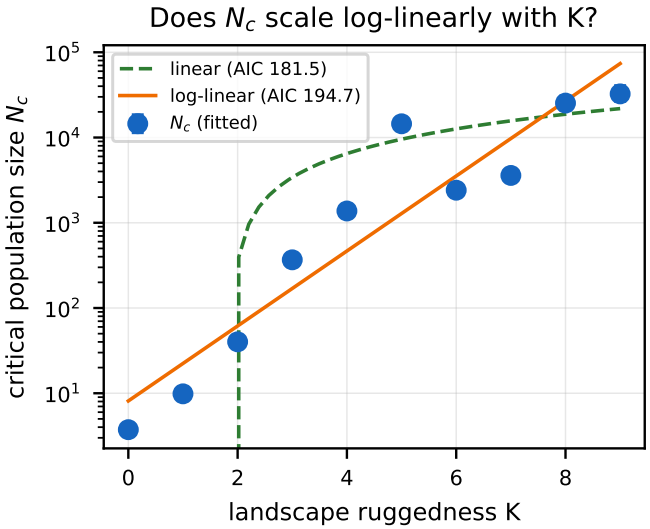


Figure 2: $N_c(K)$ with linear and log-linear fits overlaid (log scale).

Table 2: Measured $s(K)$ and $N_c^{\text{Kimura}}(K) = 1/s(K)$.

K	$s(K)$	K	$s(K)$
0	0.0336	5	0.0884
1	0.0519	6	0.0938
2	0.0583	7	0.1017
3	0.0691	8	0.1066
4	0.0776	9	0.1149

4.3 Effective population size

N_e, estimated from allele-frequency trajectories at a fixed census size $N = 500$ with no elitism, itself increases with K (roughly 100 at $K = 0$ to 180 at $K = 9$) – the same order of magnitude as N throughout, confirming the GA remains a reasonable Wright-Fisher-like object to compare against Kimura's formula at every K tested, not just at $K = 0$.

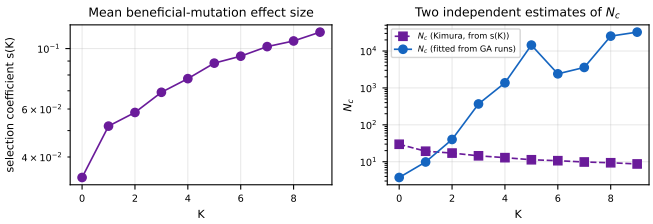


Figure 3: Left: measured selection coefficient $s(K)$, monotonically increasing. Right: the two independent $N_c(K)$ estimates, moving in opposite directions.

5 Discussion

The population-size threshold is not the surprising part of this result – that bigger populations help on harder landscapes is close to the null expectation. The surprising part is that the specific, quantitative, textbook mechanism for *why* – shrinking per-mutation selection strength letting drift dominate, exactly as Kimura's formula describes for a single locus in a real population – points the wrong way. NK-model epistasis does not appear to dilute the effect of a single beneficial mutation; if anything, in landscapes with more interacting loci, a bit flip that does help tends to help by more, not less, on average.

A more plausible mechanism, consistent with what is already known about NK landscapes – that the number of local optima grows with K [2, 6] – is that the binding constraint at high K is not single-locus fixation probability but population-level exploration: a small population converges onto one local optimum's basin before crossover and mutation can discover a better one elsewhere, while a larger population maintains enough simultaneous diversity to explore multiple basins in parallel. This is a population-*diversity* argument, not a population-*genetics* one in the classical single-locus sense, and it was not tested directly here (it would require tracking, e.g., how many distinct basins of attraction a population's genomes span over time). We flag it as the natural next hypothesis rather than claim it is confirmed.

A second limitation is resolution: the local pilot's 4-point K grid gave a genuinely weak test of linear-vs-log-linear scaling. The GPU-scale run's 10-point grid is a materially stronger test of the same question, and it reaches the same conclusion (linear preferred by AIC) – so this is not merely "the pilot was underpowered and the real answer is log-linear after all," in the spirit of this repository's project 06 pilot, but a result that held up under more scrutiny, not one that dissolved under it. The non-monotonicity in the fitted $N_c(K)$ at high K, however, did not go away with more landscape averaging (10 draws per K at GPU scale vs 5 locally), which suggests either landscape-instance variance at high K needs substantially more than 10 draws to average out,

or the true $N_c(K)$ relationship itself is not smooth – we cannot distinguish these with the current data and report the non-monotonicity as-is rather than picking whichever explanation is more convenient.

6 Conclusion

A genetic algorithm on NK fitness landscapes needs a substantially larger population to reliably reach the global optimum as epistasis increases – a real, sharp, population-size threshold. But the population-genetics mechanism proposed to explain it, a single-locus selection coefficient shrinking under epistasis until drift dominates (Kimura's classic fixation-probability crossover), is falsified by direct measurement: the selection coefficient measured from each landscape's distribution of fitness effects grows, not shrinks, with epistasis, so the textbook formula predicts the opposite trend from what is observed. Genetic algorithms may be named for evolution, but at least this piece of their population-sizing behaviour is not explained by evolution's own single-locus theory of drift and selection – a multi-peak exploration argument is proposed as the more likely mechanism, left for future work to test directly.

References

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